qiskit_device_benchmarking/bench_code/mrb/mirror_rb_experiment.py

qiskit_device_benchmarking/utilities/sampling_utils.py

* sampling_utils.py
* Matching either edge graph (networkx random weights)

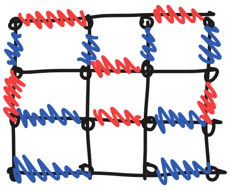
bench_code/mrb/mirror_rb_experiment.py (without QA) vs mirror_qa.py

initial_~~entangling~~_angle

Main template ⇐ 0.0 (mrb)

Actual difference happens. ⇐ pi/2 (~~omega~~)

Lattice that I can pair : Various ways that I can pair.

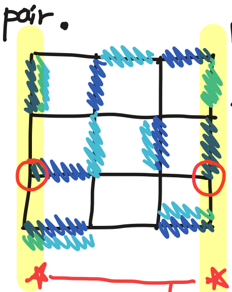


↗ Made a loop.

(If I replace all blue with red, that's another pair)

(Everything paired up)

'Square lattice'



1. Pair things

2. Choose ones that we don't want. (0)

● : pairing exept those things

(Almost everything paired up, lost pairing & finding the qubit which is abandoned.)

Information that we recover →

Always on the either or both sides.

What's the

e.g. Extreme case

$\boxed{\text{Mutual information}}$

: Abandoned qubits are either this side or that side ⇒ Information that we care : Whether there are abandoned qubits are not.

Error rate (small errors ⇒ confusing about the whereabouts)

‖

(local errors, in the small area)

Topological approach

⇒ "Topologically protected" information

Sampling method is matching density

* 'Sampling method' (IBM H/W) is edge graph by default. 'density' is not 100%

(Our topological approach goal)

① Sampling method is matching density = 100%

② Use the square lattice

① Randomly select edges (making pairs)

↓

Density : 50% (flipped — actual algorithm)

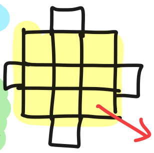
build

④-1 Sampling options (Give the random choice of leave two things either side)

④-2 We decide which ones we will abandon

③ Q.A. on square lattice after randomly selected pairs ←

② Matching method that we wrote + (Annoying) Dearrange the qubit num,



1. Use only these qubits.

(There're issues probably appear)

(Telling what I'll include ⇐ (When I make QuantumAwesomeness obj)

This is not about peak of doom.

50 samples : no abandoned

50 samples : abandoned

use whatever information

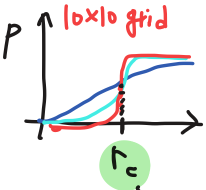
⇒ try to make the good pair as we can.

Step 1

≒ Mutual information of correct pairs vs incorrect pairs

(run 50 times) --- Run the bot which tries to solve the puzzle

Step 2



Critical round number will be there

between ~~when~~ things ~~are~~ good ~~W~~ things are have been bad. and gone

phase transitions

(Happening in the local level)

(non-local, topological, global aspect of device)

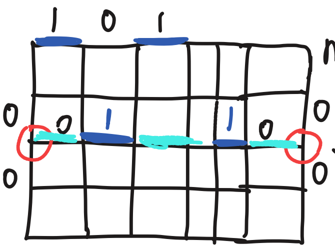
This is the one that we want to include in the paper.

c.f. Rigetti ⇒ Square lattice

1. Running on simulation

2. Bot on simulation

3. Running on Rigetti H/W



networkx sampler method for matching algorithm

mutual information → find the weights ~~etc~~. which is maximum.

Because it tries to find the best possible pairs.